CSC0056 Homework 4

- Submit your work to Moodle before 9AM, December 7th (**Monday**)
- A copy of this instruction can be found on
  - https://github.com/wangc86/csc0056
  - https://github.com/wangc86/mosquitto/tree/master/csc0056-hw4

Table of Contents:

CSC0056 Homework 4

1. Analysis of The Slotted Aloha Protocol (30 points)
2. Literature Reading (30 points)
3. Empirical Evaluation of Little's Theorem (40 points)
   - 3.1 Setting up customized publishers/subscribers (20 points)
   - 3.2 Evaluating Little's Theorem (20 points)

This homework include three parts:

1. Analysis of The Slotted Aloha Protocol (30 points);
2. Literature Reading (30 points);
3. Empirical Evaluation of Little's Theorem (40 points).

Things to submit to complete this homework assignment: all those questions/tasks marked in **boldface** in the following (which account for 100 points in total).

1. Analysis of The Slotted Aloha Protocol (30 points)

Refer to both pages 279-280 in the textbook and this lecture note. With the *no-buffering assumption* (defined in Assumption 6a in Section 4.6.1 in the textbook), and suppose that $m = 3$ and $\lambda = 1.5$ packets/second. Answer the following two questions:

1. **(15 points)** What is the average delay for a packet if $q_r = 0.2$?
2. **(15 points)** Following above, what is the average delay for a packet if $q_r = 0.8$?

2. Literature Reading (30 points)

To learn recent findings in computer science and engineering, it is very important to read research papers. Throughout the rest of this semester, we will gradually learn some tips to effectively read research papers. For this homework, do the following:

1. Read this short article on "how to read a research paper":

   > *S. Keshav. 2007. How to read a paper. SIGCOMM Comput. Commun. Rev. 37, 3 (July 2007), 83–84. DOI:https://doi.org/10.1145/1273445.1273458*

   From the link above, click the "eReader" or "PDF" to read it. Note that you may need to use the campus network to access this paper.

2. Use the strategy from the above article and read the following paper. For this homework, you only need to read the abstract, the introduction, and the concluding remarks:

> C. Wang, C. Gill and C. Lu, "Adaptive Data Replication in Real-Time Reliable Edge Computing for Internet of Things," 2020 IEEE/ACM Fifth International Conference on Internet-of-Things Design and Implementation (IoTDI), Sydney, Australia, 2020, pp. 128-134, doi: https://doi.org/10.1109/IoTDI49375.2020.00019.

Alternatively, you may download the author copy from here.

Now, answer these two questions:

1. (**10 points**) What are the challenge(s) that this paper trying to tackle?
2. (**10 points**) What are the contribution(s) that this paper claims to give?

3. (**10 points**) Write to let us know where you found difficult in reading a research paper. Your opinion will help us identify what might help students the most, and they will help us shape materials regarding paper reading in the later part of this semester.

3. Empirical Evaluation of Little's Theorem (40 points)

For this part, we will follow homework 3 and use Mosquitto as our sandbox for empirical valuation.

3.1 Setting up customized publishers/subscribers (20 points)

First of all, pull the latest version of our Mosquitto repository. At your mosquitto directory, type

`$ git pull`

which should download everything necessary for this part of the homework.

Now, different from Homework 3, we are going to compile and use our customized mosquitto publishers/subscribers. The code for both publisher and subscriber is located in folder *client*.

At folder *mosquitto*, type `make` to compile both broker, publisher, and subscriber.

Our compilation will produce a shared library *libmosquitto.so.1* in folder *lib*. This shared library will be used by both our publisher and subscriber at runtime, and we need to make it known by them. We can achieve this by updating our system's environment variable *LD_LIBRARY_PATH* to include the path to the library. This update will only apply to the current user and thus will not mess up the system-wide setting. Type the following to modify your bash configuration:

`$ vim ~/.bashrc`

And the add the following line at the end of the file:

`LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/home/cw2/repos/mosquitto/lib/; export LD_LIBRARY_PATH`

Note that the above should be placed in only one single line.

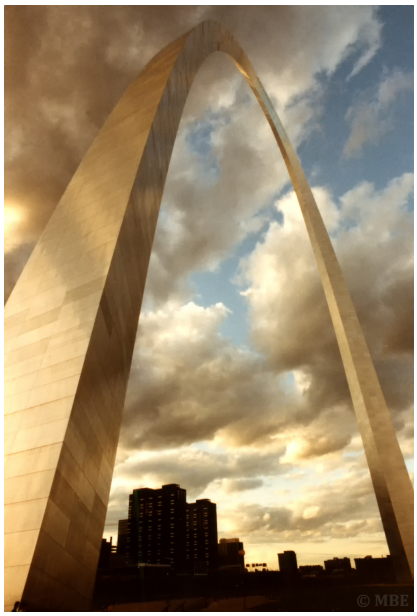
Save the change and close the file. Then type the following to have our update came into effect:

`$ source ~/.bashrc`

(20 points) Now run the script named `test-run.sh`. Take a screenshot of the output and upload it to Moodle. You should get an output similar to the following:

(The rest of this subsection is optional)

Interestingly, we may send an image using Mosquitto as a "message". For example, try sending the following image:



(The above image is from Wikimedia: https://commons.wikimedia.org/wiki/File:St._Louis_Arch_(1984).jpg. This is the landmark of the dear city where I've sojourned for seven years.)

There is a helper script named `sendImage.sh` for you to try it out. Run the script and a mosquitto subscriber will receive this image and dump it into a file named `output.jpg` :)

3.2 Evaluating Little's Theorem (20 points)

Now we are ready to evaluate